Liu, Chapter 8: Birational geometry of surfaces

Solutions and corrections to the exercises

Conventions. The 54 labels follow the three lists on printed pp. 331–332, 344–347 and 367–374 of the supplied text. A blow-up is $\mathrm{Proj}_X \bigoplus_{n\geq 0} \mathcal{I}^n$, with $\mathcal{I}\mathcal{O}_{\widetilde{X}} = \mathcal{O}_{\widetilde{X}}(1)$ and exceptional ideal $\mathcal{O}(-E)$. Arithmetic surface means a regular integral projective flat surface over a one-dimensional Dedekind base; its fibres need not be smooth or geometrically connected. Normality is used in the local-ring sense unless irreducibility is specified; where Liu additionally requires irreducibility, this distinction is stated explicitly. All divisor degrees on curves include residue-field degrees. We use the chapter's dimension formula, Serre criterion, contraction theorem and resolution theorem, not an assumed general resolution theorem in arbitrary dimension.

Corrections and proof scopes. The converse of 1.7 fails over a general affine base with non-trivial Picard group; 1.8 needs projectivity. The converse in 2.3(b) is false. In 2.12 geometric separability is unnecessarily strong over an imperfect field; the correct generic criterion is no repeated irreducible factor. In 2.9 the depth argument is for finite modules; 2.11(b) needs the base local dimension, not the dimension of the fibre. Exercise 2.14(b) is proved for every Noetherian integral X, using a local depth argument. Its codimension statement uses the infimum convention of Definition 2.5.7; purity of the base support is not asserted without additional hypotheses. In 2.15 one must exclude components mapping into proper closed subsets of the base. Exercise 3.6(a) has rank three, and the Proj formulas require section algebras rather than tensor algebras. Connected etale base change in 3.10 need not preserve integrality, and reduced in 3.13 must be strengthened to geometrically reduced. Exercise 3.14(c) has an incorrect reference to the initial normalization. Exercise 3.16(b) needs a nontrivial restricted valuation. Exercise 3.20 is false, as shown by a ruled-surface example. Exercise 3.26 fails for inseparable extensions over a non-Nagata DVR. Finally, the particular resolution in 3.27 is not a universal minimal resolution; an explicit arithmetic-surface counterexample is given. Each affected solution states exactly what is proved. In 3.6(c) we explicitly use the standard theorem on hyperelliptic families; in the corrected common-refinement statement of 3.27 we use factorization of birational maps of regular surfaces.

8.1 Blowing-ups

Exercise 1.1

> **Problem.** Show that the underlying set of the strict transform of a closed subscheme W under $\pi : \widetilde{X} \to X$, blown up in $\mathcal{I}$, is the closure of $\pi^{-1}(W \setminus V(\mathcal{I}))$.

Solution. The strict transform is the blow-up of W in the image ideal $\mathcal{J} = \mathcal{I}\mathcal{O}_W$, embedded in $\widetilde{X}$. On a chart generated by $a \in \mathcal{J}$, its ring is the subring $B[\mathcal{J}/a] \subseteq B_a$, where B is the ring of an affine piece of W; this description includes the saturation by a-power torsion. Thus a is a nonzerodivisor on every nonempty such chart. The open $D(a)$ is dense there: a nonzerodivisor belongs to no minimal prime. It lies over $W \setminus V(\mathcal{J})$. Hence this inverse image is dense in the strict transform. The latter is closed in $\widetilde{X}$, and off the center it is exactly $W \setminus V(\mathcal{I})$. Taking closures proves the assertion, even if W has components or embedded structure supported in the center; such parts are removed by saturation. □

Exercise 1.2

> **Problem.** For an invertible ideal $\mathcal{L} \subseteq \mathcal{O}_X$, prove that $\mathcal{I}$ and $\mathcal{L}\mathcal{I}$ define the same blow-up.

Solution. Multiplication identifies $(\mathcal{L}\mathcal{I})^n$ with $\mathcal{L}^{\otimes n} \otimes \mathcal{I}^n$. Locally a generator ℓ of the invertible ideal is a nonzerodivisor; division by ℓ^n identifies the degree-n pieces of their Rees algebras. On overlaps these identifications differ by multiplication by the nth power of a unit, which acts trivially on the degree-zero localization defining Proj. Thus the local Proj isomorphisms glue over X. Equivalently, twisting every graded piece by the corresponding power of an invertible sheaf does not change relative Proj. □

Exercise 1.3

> **Problem.** If a coherent ideal $\mathcal{I}$ on Noetherian X is globally generated, prove that its blow-up is projective over X.

Solution. Quasi-compactness and coherence let us select finitely many global generators, giving $\mathcal{O}_X^{r+1} \twoheadrightarrow \mathcal{I}$. Multiplication gives a surjection of graded algebras $\mathcal{O}_X[T_0, \dots, T_r] \twoheadrightarrow \bigoplus_n \mathcal{I}^n$. Relative Proj turns it into a closed immersion of the blow-up into $\mathbf{P}^r_X$. Its composite with the projection is the desired projective morphism. □

Exercise 1.4

> **Problem.** Extend Theorem 8.1.19(b),(c) to a regular closed immersion $Y \hookrightarrow X$, without requiring either scheme regular: compute the fibres, exceptional divisor and its relative canonical sheaf.

Solution. For an ideal locally generated by a regular sequence, the regular-sequence graded-algebra theorem gives

$$\operatorname{gr}_{\mathcal{I}} \mathcal{O}_X \simeq \operatorname{Sym}_{\mathcal{O}_Y}(\mathcal{I}/\mathcal{I}^2).$$

This follows by induction on the regular sequence, or the Koszul relations: the degree-n monomials in its generators are a basis modulo $\mathcal{I}^{n+1}$. Therefore

$$E = \widetilde{X} \times_X Y \simeq \operatorname{Proj}_Y \operatorname{Sym}(\mathcal{I}/\mathcal{I}^2).$$

If the conormal rank is r, its fibre at y is $\mathbf{P}^{r-1}_{k(y)}$. The exceptional ideal $\mathcal{J} = \mathcal{I}\mathcal{O}_{\widetilde{X}}$ is invertible and generated locally by a nonzerodivisor, so $E \hookrightarrow \widetilde{X}$ is a regular immersion of codimension one. Moreover $\mathcal{J}/\mathcal{J}^2 = \mathcal{O}_E(1)$; adjunction for a regular immersion gives $\omega_{E/\widetilde{X}} = \mathcal{O}_E(-1)$. If the conormal bundle is globally free, the displayed projective bundle is $\mathbf{P}^{r-1}_Y$. None of this asserts regularity of $\widetilde{X}$ for a singular ambient scheme. □

Exercise 1.5

> **Problem.** For connected projective X/k and closed points x_1, x_2, prove that a connected curve in X passes through both: reduce to the irreducible case, blow up the two points, cut by an ample divisor, and induct on dimension.

Solution. Replace X by its reduction. The finite incidence graph of its irreducible components is connected. A chain of components joining those containing the prescribed points lets us reduce to two points on a single integral component, by choosing a closed intersection point at each step and later taking the union of the resulting curves. Components of dimension zero are isolated if they occur as maximal irreducible components of a connected positive-dimensional scheme; the zero-dimensional singleton case requires a separate convention: no one-dimensional curve exists in a point. Thus assume $\dim X \geq 1$; in dimension one the reduced X itself works.

For integral X of dimension $d \geq 2$, its blow-up in the two reduced points is integral, hence connected and projective. Each exceptional fibre has positive dimension: the analytic spread of the maximal ideal of a d-dimensional local ring is d, so the fibre is $\operatorname{Proj} \operatorname{gr}_{\mathfrak{m}} A$ of dimension $d-1$. Choose an effective ample Cartier divisor on the blow-up. Its support Z is connected by the ample-divisor connectedness theorem on an integral projective variety, meets each positive-dimensional exceptional fibre, and has dimension at most $d-1$. Its proper image in X is connected, contains both points, and has smaller dimension. If that image is zero-dimensional it is a single closed point; otherwise apply the component-chain reduction and induction to it. For distinct points the zero-dimensional case cannot occur. Taking reduced images and unions gives the required connected curve in X. □

Exercise 1.6

> **Problem.** On $X = \operatorname{Spec} k[t,s]$ compare the blow-ups in (t,s) and (t^2,s). Show there is no X-morphism in either direction.

Solution. A morphism to a blow-up would make the target's center ideal invertible. On the t-chart of the first blow-up write $s = tu$. The ideal (t^2, s) becomes $t(t,u)$, not invertible at $(t,u) = (0,0)$: the maximal ideal of this regular two-dimensional local ring needs two generators. Thus the first blow-up does not map to the second.

On the s-chart of the second blow-up the ring is $B = k[t,s,v]/(sv - t^2)$. At its origin the ideal (t,s) is not principal. Indeed its images in $(t,s)/\mathfrak{m}_B(t,s)$ are the two independent classes of t, s, since the defining equation has no linear term and $\mathfrak{m}_B(t,s)$ consists of terms of order at least two. An invertible ideal over a local domain is principal, so this ideal is not invertible. Thus there is no morphism in the reverse direction either. This also proves the blow-ups are not isomorphic over X. □

Exercise 1.7

> **Problem.** For projective X/A and globally generated invertible $\mathcal{L}$, let $f : X \to \mathbf{P}^n_A$ be given by generators s_i, with scheme-theoretic image W. (a) Prove W finite over A if $\mathcal{L}$ is trivial. (b) If $W = \operatorname{Spec} B$ is finite, examine its proposed section in $\mathbf{P}^n_B$, global coordinates b_i, and the claimed trivialization of $\mathcal{L}$.

Solution. (a) Write $s_i = ea_i$ in a global frame, with $a_i \in C = H^0(X, \mathcal{O}_X)$. Properness makes C a finite A-algebra. The a_i generate C: if their ideal were proper, a point of $\operatorname{Spec} C$ over that ideal would have a preimage in X, by surjectivity of the Stein morphism, contradicting generation on X. They therefore define $\operatorname{Spec} C \to \mathbf{P}^n_A$, through which f factors. This map is proper and quasi-finite, since C/A is finite; hence finite. Its scheme-theoretic image, and therefore W, is finite over A.

(b1) Write $f_W : X \to W$. The embedding $j : W \hookrightarrow \mathbf{P}^n_A$ gives by base change a section $\sigma : W \to \mathbf{P}^n_B$. With the same generating sections over B, $g = \sigma \circ f_W$; the print reverses this composition. Scheme-theoretic image commutes with this identification, so $g(X) = \sigma(W)$.

Parts (b2),(b3) are false over a general A. Let A be a Dedekind domain with nontrivial ideal class, and choose a nonprincipal invertible ideal M. Take $X = \operatorname{Spec} A$, which is projective over itself, and $\mathcal{L} = \widetilde{M}$, generated by finitely many elements. Its map to projective space is a section, so its image is finite, yet $\mathcal{L}$ is not trivial. Global homogeneous coordinates $b_i \in B$ as in the exercise would trivialize $j^*\mathcal{O}(1)$, which is precisely the missing condition.

If $\operatorname{Pic}(B) = 0$, for instance if B is semilocal, choose a frame of that invertible B-module and write its generating sections as b_i. Then the b_i generate the unit ideal and give exactly the stated affine coordinate maps. On X_{s_i} the correct frame is s_i/b_i, not the printed product $s_i b_i$; these quotients agree on overlaps and trivialize $\mathcal{L}$. This proves the intended converse under that additional hypothesis. □

Exercise 1.8

> **Problem.** Extend the blow-up description of a projective birational map to reduced schemes: for reduced quasi-projective X over an affine Noetherian base and reduced Z, use the generic points and an ample twist to clear the denominators of $f_*\mathcal{L}$.

Solution. Projectivity of $f : Z \to X$ is essential and is omitted in the exercise's statement. A dense proper open immersion is birational and is not proper, whereas a blow-up is proper. For example $\mathbf{G}_m \hookrightarrow \mathbf{A}^1$ disproves the unrestricted statement.

Assume f projective. Choose a relatively very ample $\mathcal{L}$ and a sufficiently high power so that $Z = \operatorname{Proj} \bigoplus f_*\mathcal{L}^n$ with multiplication generated in degree one, as in Thm. 8.1.24. A rational frame on each reduced generic component embeds $\mathcal{L}$ into the total fraction sheaf of Z, and $\mathcal{F} = f_*\mathcal{L}$ into $\mathcal{K}_X$. The coherent denominator ideal $\mathcal{J} = \operatorname{Ann}((\mathcal{F} + \mathcal{O}_X)/\mathcal{O}_X)$ is the unit ideal at all generic points.

(a) Remove component intersections and shrink about the finitely many generic points to get disjoint affine neighborhoods U_i on which a regular section $s \in \mathcal{J}(U)$ exists, $U = \coprod U_i$. In fact one can shrink until $\mathcal{J}|_U = \mathcal{O}_U$ and take $s = 1$.

(b) In a projective embedding of X, homogeneous prime avoidance supplies a principal projective open containing these generic points and intersecting X inside U. This is the hypersurface avoidance lemma used in the question; it works over a finite base as well.

(c) Denote its defining section by h. Since $\mathcal{J}$ is the unit ideal on this principal open, the extension-by-an-ample-power lemma extends a sufficiently high power of h to a global section of $\mathcal{J}(d)$, still nonzero at every generic point. On reduced X it is a regular section. Thus it gives an injection $\mathcal{O}_X(-d) \hookrightarrow \mathcal{J}$. Multiplying $\mathcal{F}$ by its image gives a coherent ideal $\mathcal{I} \subset \mathcal{O}_X$. Its powers are the twisted degree pieces of $\bigoplus f_*\mathcal{L}^n$. Twisting a graded algebra by powers of an invertible sheaf preserves Proj, so $Z = \operatorname{Bl}_{\mathcal{I}} X$. This is exactly the integral proof with regularity at all generic components in place of nonzero at one generic point. □

8.2 Excellent schemes

Exercise 2.1

> **Problem.** For one-dimensional Dedekind domains $A \subseteq B \subseteq K = \operatorname{Frac} A$, (a) prove $B_{\mathfrak{q}} = A_{\mathfrak{q}\cap A}$ for maximal $\mathfrak{q}$; (b) deduce injectivity on spectra and an open immersion when A is semilocal; (c) express B as the intersection of the local rings at image points.

Solution. For $0 \neq b \in \mathfrak{q}$, write $b = a/c$ with $a, c \in A$, $c \neq 0$. Then $0 \neq a = cb \in \mathfrak{q} \cap A$, so its contraction $\mathfrak{p}$ is a maximal ideal. The inclusion $A_{\mathfrak{p}} \subseteq B_{\mathfrak{q}}$ is local and both are DVRs with the same fraction field. A valuation ring has no proper local overring in the same field dominating it: any element outside it has inverse in its maximal ideal and cannot belong to the dominating ring. Thus the rings are equal.

Their maximal ideals recover $\mathfrak{q}$ by contraction to B, so there is at most one point above each $\mathfrak{p}$. The generic point maps to the generic point, proving injectivity. If A is semilocal the image, the generic point plus a subset of finitely many closed points, is open. The map is a homeomorphism onto that open, and (a) gives isomorphisms on stalks; hence it is an open immersion. Finally $B = \bigcap_{\mathfrak{q} \in \operatorname{Max} B} B_{\mathfrak{q}}$ in its fraction field, giving $B = \bigcap_{\mathfrak{p} \in F} A_{\mathfrak{p}}$. Including the generic point in that intersection does not change it. □

Exercise 2.2

> **Problem.** Let Z be catenary locally Noetherian. Suppose the dimension formula is an equality for every dominant locally finite type map to every integral closed subscheme of Z. Show Z universally catenary.

Solution. Work on $\operatorname{Spec} A \subset Z$ and in $A[T_1, \dots, T_n]$. For primes $P \subset Q$, put $p = P \cap A$, $q = Q \cap A$. Apply the assumed formula to the integral scheme $\operatorname{Spec}(A[T]/P) \to \operatorname{Spec}(A/p)$ at Q/P. It gives

$$\operatorname{ht}(Q/P) = \operatorname{ht}(q/p) + \operatorname{trdeg}_{\kappa(p)} \kappa(P) - \operatorname{trdeg}_{\kappa(q)} \kappa(Q).$$

The same equation applies to any intermediate prime H, using its contraction to A. The two residue transcendence terms telescope when adding the equations for $P \subset H$ and $H \subset Q$, and catenarity of A adds the two base heights. Thus $\operatorname{ht}(Q/P) = \operatorname{ht}(H/P) + \operatorname{ht}(Q/H)$. Polynomial rings over A are catenary, and so are their quotients, which are all finite type A-algebras. This is universal catenarity. The formula is understood after localizing at the indicated primes, so the required maps are covered by the hypothesis. □

Exercise 2.3

> **Problem.** For integral locally Noetherian X blown up in a proper closed subscheme Z, (a) show finiteness of the blow-up forces every component of Z to have codimension one; (b) examine the claimed converse in dimension two.

Solution. Assume the ideal is nonzero, so the blow-up is birational and surjective. The exceptional preimage is an effective Cartier divisor, hence all its irreducible components have codimension one. If the map is finite, a generic point ξ of Z is the image of a generic point of this preimage. Here is a local dimension argument which does not assume going-down for a nonnormal target. Put $A = \mathcal{O}_{X,\xi}$; the localized finite blow-up has a finite integral algebra B over A. Since ξ is minimal in the center, $V(IB)$ consists only of the finitely many maximal ideals above $\mathfrak{m}_A$. At each of them the nonzero invertible ideal IB is principal and its radical is that maximal ideal. The principal ideal theorem makes every such local ring one-dimensional. Integral dimension equality gives $\dim A = \dim B = 1$, proving (a).

The converse is false, even for a regular surface. On $X = \operatorname{Spec} k[x, y]$ take $I = x(x, y) = (x^2, xy)$. The center has the single irreducible component $V(x)$ of codimension one. But 1.2 identifies its blow-up with the blow-up of (x, y), which has a fibre $\mathbf{P}^1$ over the origin and is not finite. A correct sufficient condition is that the center ideal be invertible; then its blow-up is the identity. Merely specifying the codimensions of its components does not give that condition. □

Exercise 2.4

Problem. For Noetherian local A and a regular $t \in \mathfrak{m}_A$, show that Cohen–Macaulayness of A/tA implies that of A.

Solution. The depth-drop theorem gives $\operatorname{depth} A = 1 + \operatorname{depth}(A/tA)$. A nonzerodivisor in the maximal ideal lowers the dimension by one, so $\dim A = 1 + \dim(A/tA)$. Equality of depth and dimension for the quotient therefore gives the same equality for A. □

Exercise 2.5

Problem. Prove polynomial rings over a Noetherian Cohen–Macaulay ring are Cohen–Macaulay, and deduce universal catenarity.

Solution. Localize a polynomial ring at Q and put $p = Q \cap A$. The map $A_p \to A[T]_Q$ is flat local, its closed fibre is a localization of a polynomial ring over $\kappa(p)$, hence regular. The flat local depth formula proved in 2.9, with the flat local dimension formula, gives equality of depth and dimension. Thus every such localization is Cohen–Macaulay. By Prop. 8.2.15, Cohen–Macaulay local rings are catenary. Consequently every polynomial ring over A is catenary, and so are its quotients, proving universal catenarity. This use of 2.9 is forward reference to its independent regular-sequence proof below. □

Exercise 2.6

Problem. For a nonzero finite module M over Noetherian local $(A, \mathfrak{m})$, prove depth zero equivalent to $\mathfrak{m} \in \operatorname{Ass} M$, and to containing a copy of $A/\mathfrak{m}$.

Solution. Depth zero means every element of $\mathfrak{m}$ is a zero divisor on M. The zero divisors are the union of the finite set $\operatorname{Ass} M$, so prime avoidance forces $\mathfrak{m}$ to be one of those primes. Conversely membership of $\mathfrak{m}$ in that union makes every element of $\mathfrak{m}$ a zero divisor, so depth is zero. Finally $\mathfrak{m} = \operatorname{Ann}(m)$ for a nonzero $m \in M$ is exactly an injection $A/\mathfrak{m} \to M$, $1 \mapsto m$. □

Exercise 2.7

Problem. For a finite module M over Noetherian A, prove $\bigcup_{0 \neq x \in M} \operatorname{Ann}(Ax) = \bigcup_{p \in \operatorname{Ass} M} p$.

Solution. An element of the left side kills a nonzero element of M. The nonzero cyclic submodule it kills has an associated prime; this prime also belongs to $\operatorname{Ass} M$ and contains that element. Conversely if $p = \operatorname{Ann}(x)$ is associated, every element of p kills the nonzero cyclic module Ax. This proves both inclusions, including $M = 0$, when both unions are empty. □

Exercise 2.8

> **Problem.** (a) Prove reducedness equivalent to $(S_1)+(R_0)$. (b) Decompose the total fraction ring and normalization of a reduced Noetherian ring by its minimal primes. (c) Prove the general, not necessarily integral, Serre normality criterion.

Solution. (a) A reduced Noetherian ring has no embedded primes and has fields at its minimal primes, giving both properties. Conversely (S_1) says there are no embedded points; (R_0) says its generic local rings are fields. The nilradical, if nonzero, would have an associated point among those of the ring, all generic, contradicting this generic reducedness.

(b) If p_i are the finitely many minimal primes, localization away from their union gives the Artinian reduced ring

$$Q(A) = \prod_i \operatorname{Frac}(A/p_i).$$

The injection $A \to \prod_i A/p_i$ is finite, since it is a finite direct sum of cyclic A-modules, hence integral. An element of the product of fraction fields is integral over A exactly when it is integral over this finite intermediate ring, componentwise. Thus $A' = \prod_i (A/p_i)'$; no finiteness of these individual normalizations is asserted without Nagata hypotheses.

(c) Normality implies $(R_1), (S_2)$ componentwise. Conversely these imply (a), hence reducedness. If $a/b \in Q(A)$ is integral, with b regular, then A/bA has (S_1) by the depth-drop inequality from (S_2). At its generic points, all of height one in A, the local rings of A are DVRs by (R_1), so a/b lies in them. Therefore the class of a in A/bA vanishes at all its associated points and is zero. Hence $a/b \in A$. Thus A is integrally closed in $Q(A)$. The idempotents of its finite product of fraction fields are integral, hence belong to A; they split A into integrally closed domains. Their local rings are normal domains, proving the criterion without integrality or connectedness assumptions. □

Exercise 2.9

> **Problem.** For flat local Noetherian $A \to B$ and nonzero module M, prove $\operatorname{depth}_B(B \otimes_A M) = \operatorname{depth}_A M + \operatorname{depth}(B/\mathfrak{m}_A B)$ by reducing along regular elements in the base and in the closed fibre.

Solution. We assume M finite, as required for the finite-module depth and Nakayama arguments in this exercise. Write $C = B/\mathfrak{m}_A B$ and $N = B \otimes_A M$. If both depths on the right are zero, 2.6 gives $A/\mathfrak{m}_A \hookrightarrow M$. Flatness gives $C \hookrightarrow N$, and a copy of $k(B)$ in C then embeds in N, so its depth is zero.

If $x \in \mathfrak{m}_A$ is M-regular, flatness makes it N-regular. Passing to A/xA, B/xB, M/xM lowers both module depths by one and leaves the closed fibre unchanged. If $y \in \mathfrak{m}_B$ is regular on C, the fibrewise flatness criterion makes B/yB flat over A and y regular on B. Tensoring $0 \to B \xrightarrow{y} B \to B/yB \to 0$ with M remains exact, because the quotient is A-flat. Thus y is N-regular, and passing to B/yB lowers its depth and the fibre depth by one, while leaving M over A unchanged. The tensor product here is $(B/yB) \otimes_A M$, correcting the printed ill-typed tensor over B. Induct on the sum of the two nonnegative finite depths. The quotients remain nonzero finite modules by Nakayama, so each induction step is legitimate. □

Exercise 2.10

Problem. For flat local Noetherian $A \to B$ with B regular of dimension at most one, find $a \in \mathfrak{m}_A$ with $\mathfrak{m}_A B = aB$, deduce $\mathfrak{m}_A = aA$, and prove A regular.

Solution. A flat local map is faithfully flat and injective. If B is a field, locality gives $\mathfrak{m}_A = 0$. Otherwise B is a DVR. If $\mathfrak{m}_A B = 0$ again A is a field. In the remaining case, among finite generators of $\mathfrak{m}_A$ choose a with minimal positive valuation in B. Its image generates $\mathfrak{m}_A B$. Faithfully flat contraction gives $\mathfrak{m}_A = (aB) \cap A = aA$. Since A embeds in the domain B, it is a domain. A Noetherian local domain with maximal ideal generated by one nonzero element has dimension and embedding dimension one, so is a DVR and regular. □

Exercise 2.11

Problem. For surjective flat $f : X \to Y$, (a) descend normality; (b) with Cohen–Macaulay Y, compare the Serre conditions on a fibre and on X at its points; (c) characterize Cohen–Macaulayness of X by that of the base and fibres.

Solution. (a) At $x \mapsto y$ the map $A = \mathcal{O}_{Y,y} \to B = \mathcal{O}_{X,x}$ is faithfully flat. If B is normal, it is a domain, hence so is A. An element of $\operatorname{Frac} A$ integral over A lies in B by normality; faithful flatness gives $B \cap \operatorname{Frac} A = A$, using $(aB) \cap A = aA$ on a denominator. Thus A is normal.

(b) The correct shift is $a = \dim \mathcal{O}_{Y,y}$, not $d = \dim X_y$. At each $x \in X_y$, put $C = \mathcal{O}_{X_y,x}$. The depth and dimension formulas give

$$\operatorname{depth} B = a + \operatorname{depth} C, \qquad \dim B = a + \dim C.$$

Thus $\operatorname{depth} C \geq \min(k, \dim C)$ is equivalent to $\operatorname{depth} B \geq \min(a + k, \dim B)$. This proves the corrected equivalence simultaneously at every point of the fibre. The printed shift is false: over $\operatorname{Spec} k$, take the reduced union of two affine planes meeting only at the origin. It has (S_1) but not (S_2) there (its local depth is one), whereas the printed $d + k$ with $d = 2, k = 1$ would require Cohen–Macaulayness. The same corrected formulas with k large give (c): equality of depth and dimension upstairs holds exactly when it holds for the base and the closed fibre. Surjectivity supplies a point over each base point.

For Lemma 4.1.18, the base is a DVR and the closed fibre is reduced, hence (S_1). At a special-fibre point with positive-dimensional fibre local ring, the depth formula gives depth at least two; at a generic special-fibre point it gives depth at least one and total local dimension one. Thus the total space has (S_2) along the special fibre, and it has (S_2) off it by generic normality. At a height-one special-fibre point the reduced zero-dimensional quotient by the uniformizer is a field, so the maximal ideal is generated by that uniformizer and the local ring is a DVR. This proves (R_1), and Serre's criterion proves the lemma. □

Exercise 2.12

> **Problem.** For a Dedekind domain R with 2 invertible and $P \in R[x]$, prove normality of $R[x,y]/(y^2 - P(x))$ equivalent to squarefreeness of P over $K = \operatorname{Frac} R$ and radicality of its coefficient-content ideal.

Solution. The correct generic condition over an arbitrary field $K = \operatorname{Frac} R$ is that P have no repeated *irreducible factor in* $K[x]$. This is weaker than having no repeated geometric root when K is imperfect. Exclude $P = 0$. The hypersurface is flat over R and Cohen–Macaulay, hence has (S_2). At a generic-fibre point with $y \neq 0$, the derivative $2y$ gives regularity. At a point $y = f(x) = 0$ for an irreducible factor f of P, the ambient regular local ring has parameters y, f. The equation $y^2 - P$ is outside their square exactly when the factor f occurs once. This proves the stated corrected generic criterion without assuming perfectness.

At a prime π of R, if $\bar{P} \neq 0$, $y^2 - \bar{P}$ is reduced over $\kappa(\pi)(x)$, because 2 is invertible. The height-one local rings on that fibre are regular: a uniformizer cuts a reduced zero-dimensional quotient, hence a field locally. If all coefficients are divisible by π, the sole generic special-fibre point has ambient local parameters π, y. Its equation is outside their square exactly when the least coefficient valuation is one. If that valuation is at least two the total local ring is not regular. Thus the vertical (R_1) condition is precisely radicality of the coefficient-content ideal. Together with the generic criterion this proves normality by Serre.

To see why literal geometric separability is not necessary, let $k = \mathbf{F}_p(a)$ with odd p, $R = k[\pi]_{(\pi)}$, and $P = x^p - a$. It is irreducible over K but has derivative zero. The resulting ring is nevertheless regular: using $a = x^p - y^2$, it is a localization of the polynomial ring $\mathbf{F}_p[x, y, \pi]$. Its content ideal is the unit ideal. Thus the printed "without multiple root" equivalence needs the above factor-based interpretation, or a perfect fraction field. □

Exercise 2.13

> **Problem.** Let A be regular local of dimension d, and M a finite Cohen–Macaulay module with full support. (a) Choose a minimal finite free cover. (b) Show nonzero $t \in \mathfrak{m}_A$ is M-regular and describe the quotient. (c) Prove the cover has zero kernel. (d) If M is also a nonzero unital algebra, include 1 in a basis.

Solution. Lift a residue-field basis to obtain $0 \to N \to L \to M \to 0$ with L finite free and $L \otimes k \simeq M \otimes k$. A full-support Cohen–Macaulay module over the regular local domain has only the minimal prime (0) as an associated prime, by unmixedness. Thus every nonzero t is M-regular. Tensoring with A/tA gives an exact sequence with first term N/tN. The quotient M/tM is Cohen–Macaulay by depth drop and has full support in A/tA, by Nakayama localized at each prime containing t.

To prove (c), choose specifically $t \in \mathfrak{m}_A \setminus \mathfrak{m}_A^2$, so A/tA is regular. Induct on d, starting with a field. The induction makes M/tM free, so the sequence modulo t splits. Tensor it with k; the map $N \otimes k \to L \otimes k$ is injective, but its image is zero by minimality of the cover. Hence $N/\mathfrak{m}_A N = 0$, and Nakayama gives $N = 0$. Finally $1 \notin \mathfrak{m}_A M$ in a nonzero finite algebra: otherwise multiplication would give $M = \mathfrak{m}_A M$, impossible by Nakayama. Extend its class to a residue basis and lift; since M is free, these lifts form a basis including 1. □

Exercise 2.14

> **Problem.** For $\pi : Y = \mathbf{A}^1_X \to X$, (a) prove injectivity of π^* on Picard groups. (b) If X is Noetherian integral, represent each $\mathcal{L} \in \mathrm{Pic}(Y)$ by a Cartier divisor D with $\operatorname{Supp} D = \pi^{-1}(E)$ for a closed subset E of codimension one. (c) Prove π^* an isomorphism when X is regular.

Solution. (a) The zero section s has $s^*\pi^* = \mathrm{id}$, proving injectivity.

(b) *A local depth observation.* Let $(R, \mathfrak{m})$ be a Noetherian local domain of depth at least two. If a Cartier divisor on $\operatorname{Spec} R$ is zero off the closed point, it is zero everywhere. Indeed choose a regular sequence $a, b \in \mathfrak{m}$. A local equation $f \in \mathrm{Frac}(R)^*$ and its inverse are regular on both $D(a)$ and $D(b)$, hence lie in $R_a \cap R_b = R$. For the last equality, write $f = u/a^n = v/b^m$; then $b^m u \in a^n R$, and regularity of b^m modulo a^n implies $u \in a^n R$. The same argument applies to f^{-1}. Thus f is a unit. Consequently the local ring at a generic point of the support of any nonzero Cartier divisor on an integral Noetherian scheme has depth at most one.

Choose a divisor with no horizontal support. Put $K = K(X)$ and let η be the generic point of X. Since $K[T]$ is a PID, $\mathcal{L}|_{Y_\eta}$ is free. Choose a generator there and regard it as a rational section of $\mathcal{L}$ on integral Y. Its divisor D_0 represents $\mathcal{L}$ and restricts to zero on the whole generic fibre. Thus $\operatorname{Supp} D_0$ misses Y_η.

Let z be a generic point of an irreducible component of this support and put $x = \pi(z)$. In particular $x \neq \eta$. The map $A = \mathcal{O}_{X,x} \to B = \mathcal{O}_{Y,z}$ is flat local. If z were not the generic point of the fibre $\mathbf{A}^1_{k(x)}$, it would be a closed point of that fibre; its fibre local ring would be a regular local ring of dimension one. Since A is a local domain which is not a field, $\operatorname{depth} A \geq 1$. The flat local depth formula of 2.9, applied to the finite A-module A, would give

$$\operatorname{depth} B = \operatorname{depth} A + \operatorname{depth} \mathcal{O}_{Y_x,z} \geq 2.$$

This contradicts the observation above: after localization at this generic support point the divisor has support only at the closed point of $\operatorname{Spec} B$. Therefore z is the generic point of Y_x.

If $x_1, \ldots, x_r$ are the finitely many resulting base points, the closure of each such z_i is exactly $\mathbf{A}^1_{\overline{\{x_i\}}}$, because that scheme is integral and z_i is its generic point. Hence

$$\operatorname{Supp} D_0 = \pi^{-1}(E_0), \qquad E_0 = \bigcup_{i=1}^{r} \overline{\{x_i\}} \subsetneq X.$$

This proves the required vertical-support assertion without normality. It also applies to any Cartier divisor whose restriction to Y_η is zero.

Arrange codimension exactly one. Definition 2.5.7 defines the codimension of a closed set as the infimum of the codimensions of its irreducible components, not as a purity assertion. If $\dim X \geq 1$, choose a codimension-one point x of X and a nonzero $h \in \mathfrak{m}_{X,x} \subset K$. Let z be the generic point of Y_x. Then $B = \mathcal{O}_{Y,z}$ is a one-dimensional local domain and

$$e = \operatorname{mult}_z(\operatorname{div}_Y(h)) = \operatorname{length}_B(B/hB) > 0.$$

Write $c = \operatorname{mult}_z D_0$ and choose a positive integer N with $c + Ne \neq 0$. The linearly equivalent divisor $D = D_0 + N \operatorname{div}_Y(h)$ still has zero generic-fibre restriction. By the preceding argument

its support is $\pi^{-1}(E)$ for a proper closed $E \subset X$. Its nonzero multiplicity at z implies $x \in E$. Therefore

$$1 \leq \operatorname{codim}(E, X) \leq \operatorname{codim}(\overline{\{x\}}, X) = 1.$$

This proves exactly the requested codimension statement. It permits components of larger codimension; no injectivity of Cartier divisors into Weil cycles was used. If X has dimension zero, it is the spectrum of a field and the line bundle is trivial; take $D = 0, E = \varnothing$. This empty-support case is the necessary degenerate exception to literal codimension one, also available whenever a trivial bundle is represented by the zero divisor.

(c) Now assume X regular and first integral. The scheme Y is regular, so Cartier and Weil divisors agree. Use D_0 above. Its nonzero prime cycles are vertical, and a height-one prime $P \subset A[T]$ with nonzero contraction p is $pA[T]$, with $\operatorname{ht} p = 1$. Indeed flat going-down bounds the height of p by one, and the prime $pA[T] \subseteq P$ already has height one. Thus the cycle is $\sum n_i[\pi^{-1}(Z_i)]$ for base prime divisors Z_i. Since X is regular, $C = \sum n_i Z_i$ is a Cartier divisor on X. At its generic points the local map to the generic fibre point of Y is an extension of DVRs with the same uniformizer and ramification index one. Therefore D_0 and π^*C have the same Weil divisor, hence are equal as Cartier divisors on regular Y. This makes $\mathcal{L}$ a pullback. Combined with (a) it proves the isomorphism. A general regular Noetherian X is the finite disjoint union of its integral components, so the result follows componentwise. □

Exercise 2.15

> **Problem.** For finite surjective $f : X \to Y$ with regular Y, (a) deduce flatness if X is Cohen–Macaulay; (b) apply to normal X when $\dim Y = 2$; (c) show the branch locus is empty or pure codimension one in the generically separable case.

Solution. Require every component of X to dominate a component of Y, for example X, Y integral. Otherwise $X = Y \amalg \{y\}$ for a closed point on a regular surface Y is finite surjective and even regular, but not flat, disproving (a). It also disproves (b) if normality is read componentwise; with Liu's additional irreducibility convention, (b) already excludes this example and is valid as written.

Under the stated correction, localize at y. A regular system of parameters of $A = \mathcal{O}_{Y,y}$ is a system of parameters in each local factor of the finite algebra $B = (f_*\mathcal{O}_X)_y$, with the same dimension, by the finite dimension formula over the universally catenary regular base. Cohen–Macaulayness makes it a regular sequence on all these local factors, hence on B. Thus B is a full-support maximal Cohen–Macaulay A-module and is free by 2.13, proving (a). In dimension two, a normal X is Cohen–Macaulay by (S_2), proving (b).

(c) On each integral base component the trace pairing is a map between locally free modules of the same rank, generically invertible by separability. Its determinant is a nonzero local equation of the discriminant. A fibre trace pairing is perfect precisely when its finite algebra is etale, so its vanishing set is exactly the branch locus. A nonzero principal equation on a regular integral scheme has all components of codimension one. This proves the assertion under the corrected flatness hypotheses, without assuming that the relative dualizing sheaf on X is invertible. □

Exercise 2.16

> **Problem.** For a PID $A = \mathcal{O}_K$, a separable quadratic extension $L/K(T)$, and the integral closure B of $A[T]$ in L, show $B/A[T]$ invertible and $B = A[T] \oplus b_0 A[T]$.

Solution. First B is finite. Choose a field basis in L consisting of integral elements; the trace pairing embeds the integral closure in the dual of their finite free $A[T]$-lattice. Traces of integral elements lie in the integrally closed ring $A[T]$, and separability makes that dual lattice finite. Since $A[T]$ is Noetherian, this proves finiteness. The normal ring B has dimension at most two and is Cohen–Macaulay. Exercise 2.15 in its dominant case makes B locally free of rank two over the regular ring $A[T]$.

The element 1 is locally part of a basis by 2.13(d), so $B/A[T]$ is locally free of rank one. By 2.14(c), $\operatorname{Pic}(A[T]) = \operatorname{Pic}(A) = 0$, since A is a PID. Choose a global generator of this quotient and lift it to $b_0 \in B$. The resulting split exact sequence gives the asserted direct sum. Here the appearances of A in the printed exercise mean $\mathcal{O}_K$. □

Exercise 2.17

> **Problem.** If X is locally of finite type over a DVR, show its regular locus is open.

Solution. Apply Lemma 8.2.37: it suffices that each integral closed subscheme contain a nonempty regular open. If that subscheme is contained in the closed fibre, it is a variety over the residue field, so has such an open by excellence of fields. If it dominates the DVR, its generic fibre is a nonempty integral variety over the fraction field and has a nonempty regular open. The generic fibre is itself open in the whole scheme, because the base generic point is $D(\pi)$, so this is a regular open of the original subscheme as well. This proves the lemma's hypothesis and the assertion. Excellence of the DVR is not required. □

Exercise 2.18

> **Problem.** For an excellent locally Noetherian scheme X, show the ring of every affine open is excellent.

Solution. Let $U = \operatorname{Spec} A \subset X$. Refine its cover by intersections with excellent affine opens to a finite principal cover $D(f_i)$ of U. Each $D(f_i)$ lies in some excellent affine $V = \operatorname{Spec} R$; its open immersion into V is quasi-compact and of finite presentation, so its affine ring A_{f_i} is a finite type R-algebra and excellent.

The three defining properties descend from this principal cover. Formal geometric regularity is tested on A_p, which equals the corresponding localization of some A_{f_i}. Polynomial algebras are catenary because catenarity is local and their principal opens above the f_i are catenary. Finally for any finite type A-algebra C, its regular locus is the union of its open regular loci on $\operatorname{Spec} C_{f_i}$, and hence is open. Thus A is universally catenary, a G-ring, and J-2, exactly excellence. □

8.3 Fibered surfaces

Exercise 3.1

> **Problem.** In $\mathbf{P}^1_{\mathbf{Z}}$, determine the closure of the generic-fibre point defined on $T_0 \neq 0$ by $t^2 - 2 = 0$.

Solution. Its scheme-theoretic closure is $V_+(T_1^2 - 2T_0^2)$. It has no point at infinity, since $T_0 = 0$ would force $T_1 = 0$. Thus it is $\operatorname{Spec} \mathbf{Z}[t]/(t^2 - 2) = \operatorname{Spec} \mathbf{Z}[\sqrt{2}]$, finite flat of degree two over $\mathbf{Z}$. At an odd prime it has two rational points when 2 is a square modulo that prime and one degree-two point otherwise. At 2 the fibre is $\operatorname{Spec} \mathbf{F}_2[t]/(t^2)$, a doubled rational point. These scheme structures distinguish the closure from merely listing its underlying closed points. □

Exercise 3.2

> **Problem.** For an automorphism of $\mathbf{P}^1_{\mathbf{Q}}$ given by a matrix in $\mathrm{PGL}_2(\mathbf{Q})$, determine the domain of its induced rational map on $\mathbf{P}^1_{\mathbf{Z}}$.

Solution. Choose a primitive integral representative $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with nonzero determinant. The rational map is

$$[T_0 : T_1] \longmapsto [aT_0 + bT_1 : cT_0 + dT_1].$$

For $p \nmid \det M$ it is everywhere defined on that fibre. For $p \mid \det M$, primitivity ensures that $M \bmod p$ has rank one, so it has exactly one projective kernel point $q_p \in \mathbf{P}^1(\mathbf{F}_p)$. The domain is $\mathbf{P}^1_{\mathbf{Z}} \setminus \{q_p : p \mid \det M\}$.

To check there is no removable indeterminacy, work over $\mathbf{Z}_{(p)}$. Row and column changes by invertible matrices put M in Smith form $\operatorname{diag}(1, p^e)$ up to units. Near its base point the generating ideal is (u, p^e) in a regular two-dimensional local ring, hence is not invertible: it has height two and two minimal generators. An extension to projective space would make the image ideal of the two sections invertible, contradicting this. Thus the displayed domain is maximal, independently of the chosen primitive representative. □

Exercise 3.3

> **Problem.** For normal flat X/R with R a DVR and uniformizer t, characterize points of the special fibre which are nonreduced in terms of component multiplicities. Prove the valuation criterion, the local reducedness assertion when all multiplicities through a point are one, and explain failure for $k[u, v]/(u^n, uv)$.

Solution. Work on one normal integral component. At a generic point of a fibre component Γ, the ambient local ring is a DVR. Writing $t = u\pi^{e_\Gamma}$ gives multiplicity $\operatorname{length}(A/tA) = e_\Gamma = \nu_\Gamma(t)$. If $e_\Gamma \geq 2$ the fibre is nonreduced at its generic point, and the support of its coherent nilradical contains the whole Γ.

Conversely let $A = \mathcal{O}_{X,x}$ and suppose the multiplicities of all fibre components through x are one. If $a^n = tb$ in A, then at each corresponding height-one valuation $n\nu_\Gamma(a) \geq 1$, so $\nu_\Gamma(a) \geq 1$ and $\nu_\Gamma(a/t) \geq 0$. At every other height-one prime of A, t is a unit and the same inequality is automatic. A normal Noetherian domain is the intersection of its height-one valuation rings, so $a/t \in A$. Thus $a \in tA$ and A/tA is reduced. This proves that the nonreduced locus is exactly the union of the components with multiplicity at least two.

Alternatively (S_2) for A gives (S_1) for A/tA. It is reduced exactly when generically reduced, and the generic condition is exactly $e_\Gamma = 1$. This proves the assertion without restricting x to a closed point. Finally $k[u,v]/(u^n, uv)$ has one generically reduced component $u = 0$ of multiplicity one, but has a nonzero nilpotent u at the origin. Its embedded associated point violates (S_1), explaining the failure without normality of the ambient construction. □

Exercise 3.4

> **Problem.** Give a fibered surface over $\operatorname{Spec} \mathbf{Z}$ with generic fibre $\mathbf{P}^1_{\mathbf{Q}}$ and an irreducible special fibre of multiplicity two.

Solution. For any prime p take $X = \operatorname{Proj} \mathbf{Z}[x,y,z]/(x^2 - pyz)$. Its homogeneous ring is a domain by irreducibility of the primitive polynomial, so it is integral and torsion-free over $\mathbf{Z}$, hence flat. It is projective of dimension two. The generic conic is smooth and has the rational point $[0:1:0]$, hence is $\mathbf{P}^1_{\mathbf{Q}}$. The fibre at p is $V_+(x^2)$, the double line with reduced support $x = 0$. At its generic point the local length is two, giving the required multiplicity. Regularity of this total space is not required by the exercise. □

Exercise 3.5

> **Problem.** For a fibered surface X/S with generic fibre $\mathbf{P}^1_{K(S)}$ and geometrically integral fibre X_s, (a) over local S prove $X_s \simeq \mathbf{P}^1_{k(s)}$; (b) prove the line bundle of a section very ample; (c) deduce X is a trivial projective line over a neighborhood of s.

Solution. Assume $S = \operatorname{Spec} R$ is local. Flat projectivity makes the arithmetic genus constant, so $p_a(X_s) = 0$. A geometrically integral genus-zero curve is a smooth conic by Prop. 7.4.1. A rational point on the generic projective line extends to an R-section by properness, giving a rational point on that conic. Thus it is $\mathbf{P}^1$.

The generic and special fibres are smooth, so flatness makes X/R smooth. A section is therefore an effective Cartier divisor D. Its restriction has degree one on each fibre, so $\mathcal{O}_X(D)$ restricts to $\mathcal{O}_{\mathbf{P}^1}(1)$. Cohomology and base change gives a free rank-two module $V = f_*\mathcal{O}_X(D)$ and a surjective evaluation map. The associated map $X \to \mathbf{P}(V)$ is an isomorphism on both fibres; the relative very-ampleness criterion makes it a closed immersion, hence an isomorphism since its defining ideal has zero fibres and is zero by Nakayama. This proves (b) and the local trivialization. For general S, this finite-presentation isomorphism over $\mathcal{O}_{S,s}$, together with its inverse, descends to an open neighborhood after clearing finitely many denominators. This proves (c). □

Exercise 3.6

> **Problem.** For a smooth arithmetic surface $\pi : X \to S$ with geometrically connected generic fibre of genus g, (a) analyze the anticanonical series when $g = 0$; (b) analyze the canonical series when $g \geq 1$ and the nonhyperelliptic embedding; (c) extend a generic hyperelliptic involution and describe its quotient.

Solution. Smooth properness makes all fibres geometrically connected with the same genus. (a) On a genus-zero geometric fibre the anticanonical bundle is $\mathcal{O}_{\mathbf{P}^1}(2)$, so its space of sections has dimension *three*, not the printed two. Cohomology and base change gives a rank-three bundle $V = \pi_* \omega_{X/S}^{\vee}$. Evaluation embeds X as a relative conic in $\mathbf{P}(V)$. If S is affine, finitely many global generators of V give a closed immersion of this projective bundle into a projective space over S, so the anticanonical bundle is relatively very ample.

The correct reconstruction is

$$X \simeq \operatorname{Proj}_S \bigoplus_{n \geq 0} \pi_* (\omega_{X/S}^{-n}).$$

The printed tensor algebra $\bigoplus V^{\otimes n}$ is generally noncommutative, so ordinary Proj is not applicable. Even replacing it by $\operatorname{Sym} V$ would give a projective plane bundle instead of its conic; the conic relation is indispensable.

(b) Fibrewise duality gives $h^0(\omega) = g$ and constant $h^1(\omega) = 1$, so $V = \pi_* \omega$ is locally free of rank g and commutes with base change. The canonical bundle is generated by its sections on every geometric fibre for $g \geq 1$, hence evaluation $\pi^* V \to \omega$ is surjective. On affine S this gives generation by finitely many global sections. If no geometric fibre is hyperelliptic and $g \geq 3$, the fibrewise canonical maps are embeddings, hence the relative one is a closed immersion. Reconstruction uses the canonical section algebra $\bigoplus_n \pi_* \omega^n$, or equivalently the symmetric algebra modulo the canonical ideal, not the tensor algebra in the print.

(c) Apply the uniqueness theorem for smooth positive-genus models, proved in 3.15 below, to extend σ and its inverse. The two extensions compose to the identity since they do so generically. We use here the standard theorem on smooth hyperelliptic families: the hyperelliptic locus is closed in a smooth proper family, and the extended canonical involution has a finite flat degree-two quotient which is a smooth genus-zero family, with formation compatible with base change. This includes characteristic two; one cannot justify it merely by averaging invariant functions. A reference is Lonsted–Kleiman, *Basics on families of hyperelliptic curves*, Section 5, in particular the equivalence of the canonical involution and a double cover of a twisted projective line. Thus $Y = X/\langle \sigma \rangle$ is smooth over S and $Y_\eta = \mathbf{P}^1_{K(S)}$. Its generic rational point extends locally over every DVR of S by properness, making each genus-zero fibre a split $\mathbf{P}^1$. The fibrewise degree-two covers are separable by the same family theorem, so all fibres of X are hyperelliptic in the restricted sense used in the book. □

Exercise 3.7

Problem. For any $f : X \to Y$, $y \in Y$, and $p : X' = X \times_Y \operatorname{Spec} \mathcal{O}_{Y,y} \to X$, (a) identify the fibre over y and (b) identify local rings at points of that fibre.

Solution. The residue field of the closed point of $\operatorname{Spec} \mathcal{O}_{Y,y}$ is $k(y)$, so associativity of fibre products gives $X'_y = X_y$. Locally write $Y = \operatorname{Spec} A$, $y = p$, and a source affine as $\operatorname{Spec} B$. The base-changed ring is $(A \setminus p)^{-1}B$. A point $x \in X_y$ corresponds to $q \subset B$ with contraction p; its local ring after this localization is still B_q, since the extra denominators already lie outside q. These local identifications give (b), including uniqueness of the point above x on the fibre. □

Exercise 3.8

Problem. For a birational S-morphism $f : X \to Y$ of normal fibered surfaces, (a) show surjectivity; (b) prove it an isomorphism if each pair of corresponding closed fibres has the same number of components; (c) deduce the assertion for a birational self-map which is a morphism; (d) examine finiteness of nonconstant self-morphisms.

Solution. Because X is proper over S and Y separated, f is proper. Its image is closed and contains the generic point, so is all of Y. The generic-fibre birational morphism is an isomorphism between normal projective curves. Each component of a closed fibre of Y has a unique strict-transform component in X, since a proper birational map to a normal scheme is an isomorphism over its codimension-one points. If the numbers of components agree, there are no additional components to contract. Any positive-dimensional fibre of f would contain such a vertical curve, so all fibres are finite. Proper and quasi-finite is finite; finite birational over normal Y is an isomorphism. This proves (a)–(c).

In (d) require the map to be nonconstant on the generic curve, equivalently dominant. Surjectivity follows again. Each target fibre component must be dominated by a source component. Since source and target are the same surface, their component counts agree, leaving no component to contract. Thus the same proper quasi-finite argument makes the map finite, without requiring degree one. If "nonconstant" means only not a constant scheme map, the printed assertion is false: $\mathbf{P}^1_S \to S \xrightarrow{\infty} \mathbf{P}^1_S$ has one-dimensional image and is nonconstant over a positive-dimensional base, but is not finite. □

Exercise 3.9

Problem. For finite $f : X \to Y$ between normal fibered surfaces and generic points $\xi \mapsto \xi'$ of closed fibre components, (a) prove divisibility of their multiplicities; (b) prove the residue-field inclusion and descend geometric reducedness or integrality of the components.

Solution. A finite map here is dominant: its closed image has the same dimension two as X, so equals integral Y. At these codimension-one points the map is a local inclusion of DVRs. If e is its ramification index and t a base uniformizer, then

$$\nu_\xi(t) = e\,\nu_{\xi'}(t).$$

By 3.3 these numbers are the fibre multiplicities, proving (a). The local ring map induces the field inclusion $k(\xi') \hookrightarrow k(\xi)$, finite because f is finite. A subfield of a separably generated field over $k(s)$ is again separably generated; if the larger extension also has no new algebraic constants, neither does the subfield. These are precisely the function-field criteria for geometric reducedness and geometric integrality of integral varieties over $k(s)$. Apply them to the two reduced component curves to prove (b). The geometric property concerns those curves, not the nilpotent multiplicity of the whole fibre. □

Exercise 3.10

> **Problem.** For an arithmetic surface X/S and connected etale $S' \to S$, examine whether $X_{S'}/S'$ is again an arithmetic surface, and give a failure for non-etale base change.

Solution. The connected etale scheme S' is regular of dimension one and integral, hence Dedekind. Base change preserves projectivity, flatness and relative dimension; $X_{S'} \to X$ is etale, so its total space is regular. However its total space need not be connected or integral.

For a counterexample to the printed integral conclusion take $S = \operatorname{Spec} k[t]$ with $\operatorname{char} k \neq 2$, a quadratic separable field extension k'/k, $S' = \operatorname{Spec} k'[t]$, and $X = \mathbf{P}^1_{S'}$ considered over S. This is an integral arithmetic surface. But $k' \otimes_k k' \simeq k' \times k'$, so $X_{S'}$ is the disjoint union of two copies of $\mathbf{P}^1_{S'}$. The correct statement is that each connected component is an arithmetic surface. The whole base change is one if, for example, the original generic fibre is geometrically integral.

Without etaleness even regularity can fail. The conic model $xy = tz^2$ over $k[t]_{(t)}$ is regular; under the ramified change $t = u^2$ its affine chart becomes $xy = u^2$, singular at $x = y = u = 0$. This provides the requested failure even when integrality is preserved. □

Exercise 3.11

> **Problem.** For an arithmetic surface X/S, prove the Zariski tangent space of any fibre at any point has dimension at most two over its residue field.

Solution. The fibre local ring is a quotient of the regular local ring $\mathcal{O}_{X,x}$ by the base uniformizer. The cotangent space of a quotient is a quotient of the original maximal ideal modulo its square, so its dimension is at most the embedding dimension of $\mathcal{O}_{X,x}$, namely its dimension, at most two. Dualizing over the residue field gives the same bound for the Zariski tangent space. This refers to the local-ring tangent space of the fibre, as in the text, not to all derivations relative to a possibly imperfect base field. □

Exercise 3.12

> **Problem.** For a closed point x on an arithmetic surface, over $s \in S$ with uniformizer t, show the fibre is regular at x exactly when $t \notin \mathfrak{m}_x^2$.

Solution. The local ring $A = \mathcal{O}_{X,x}$ is regular of dimension two, and t is a nonzerodivisor by flatness. Its quotient has dimension one and cotangent space $\mathfrak{m}_x/(\mathfrak{m}_x^2 + (t))$. This space

has dimension one exactly when the nonzero parameter class of t lowers the two-dimensional cotangent space by one, that is, $t \notin \mathfrak{m}_x^2$. Otherwise its dimension is two, exceeding the quotient's dimension, so it is not regular. □

Exercise 3.13

> **Problem.** For the constants/Stein factorization $X \to S' \to S$ of Prop. 8.3.8, assess the claim that a reduced fibre forces $S' \to S$ etale above it, and prove that smoothness of X/S implies smoothness of X/S'.

Solution. Assume X normal, as in the proposition. **Reduced must be strengthened to geometrically reduced.** Take $k = \mathbf{F}_p(a)$, $R = k[[t]]$, $R' = k(a^{1/p})[[t]]$, and $X = \mathbf{P}^1_{R'}$ over R. It is regular, flat and projective, its special fibre is reduced, and its constants factor is $\operatorname{Spec} R'$. The latter is not etale over R, because its residue extension is purely inseparable.

For a geometrically reduced fibre the constants algebra is finite etale at the base point. One proof uses degree-zero cohomology and base change for geometrically reduced proper fibres (Exercises 5.3.14(d),(e)): the fibre algebra of the Stein factor is then a product of finite separable residue extensions and the finite constants algebra is flat over the DVR. Thus the finite morphism is etale at those points. Equivalently, the multiplicity formula forces ramification index one and geometric reducedness rules out inseparable residues. If X/S is smooth its fibres are geometrically reduced, so S'/S is etale. Locally an etale base factor does not change smoothness of the remaining morphism: the graph $X \to X \times_S S'$ is an open immersion over S'. Hence X/S' is smooth. □

Exercise 3.14

> **Problem.** For Nagata Noetherian integral X and a valuation centered at x with finitely generated residue field and $\operatorname{trdeg}_{k(x)} k(\nu) = \dim \mathcal{O}_{X,x} - 1$, repeatedly blow up the reduced closure of its center. (a) Show the union of its center local rings is $\mathcal{O}_\nu$. (b) Show these local rings eventually have dimension one. (c)–(e) Use their normalizations to prove the sequence reaches the DVR $\mathcal{O}_\nu$.

Solution. First ν is discrete. Indeed the equality direction in the proof of Thm. 8.3.26 realizes a residue transcendence basis on one blow-up; finite normalization, available by the Nagata hypothesis, gives a normal center of dimension one by the dimension inequality. Its DVR is $\mathcal{O}_\nu$. This direction of that proof uses no universal-catenarity assumption. Normalize the value group to $\mathbf{Z}$; discreteness is essential for the descending-value argument that follows.

Put $A_n = \mathcal{O}_{X_n,x_n}$. These local rings form an increasing chain dominated by $\mathcal{O}_\nu$. For $f \in \mathcal{O}_\nu$, the nonzero ideal $J_n = \{b \in A_n : bf \in A_n\}$ has a finite set of generators, hence a least valuation r_n. If $r_n > 0$, choose b attaining it. Both b and bf lie in the maximal ideal, since $\nu(f) \geq 0$. On the blow-up chart dominated by ν, that maximal ideal becomes generated by an element a of least positive valuation; all ratios of its elements by a belong to A_{n+1}. Thus b/a is an admissible denominator for f and $r_{n+1} \leq r_n - \nu(a) < r_n$. Eventually $r_n = 0$, when its denominator is a unit and $f \in A_n$. This proves (a).

(b) Lift a finite set of generators of $k(\nu)$ over $k(x)$ to $\mathcal{O}_\nu$. By (a), they all lie in some A_r, so $k(x_n) = k(\nu)$ for $n \geq r$. The birational dimension inequality gives $\dim A_n \leq 1$ for these n, and the center of a nontrivial valuation cannot be generic. Thus $\dim A_n = 1$.

(c) Write B_n for the normalization of A_n, finite by Nagata. For $n \geq r$ these are semilocal Dedekind domains inside the same field, with $B_r \subseteq B_n \subseteq \mathcal{O}_\nu$. By 2.1, $\operatorname{Spec} B_n$ is an open subscheme of $\operatorname{Spec} B_r$. They form a descending chain of opens. Since a semilocal Dedekind spectrum has only finitely many points, this chain stabilizes, say at q.

Two corrections to the printed wording: the reference ring is B_r, not the normalization of the possibly higher-dimensional A_0; these open immersions need not be finite morphisms. Finiteness of the set of possible opens, not finiteness of each open immersion, is what is used.

(d) For $n \geq q$, $A_q \subseteq A_n \subseteq B_q$. The finite module B_q/A_q is torsion over the one-dimensional local domain A_q, hence has finite length. Its ascending submodules A_n/A_q stabilize. Therefore the rings A_n stabilize, say at m. (e) Their union was $\mathcal{O}_\nu$, so $A_m = \mathcal{O}_\nu$. The valuation is then of the first kind on X_m, as required. □

Exercise 3.15

> **Problem.** For smooth fibered surfaces X, Y over a one-dimensional Dedekind S, with isomorphic generic fibres of arithmetic genus at least one, prove unique extension to an S-isomorphism. Use constants reduction and point blow-ups to rule out indeterminacy.

Solution. Uniqueness follows from separatedness and agreement on the dense generic fibre. The common field of generic constants gives the same finite etale constants cover of S for both families: smooth proper fibres have an etale scheme of connected components, or use the corrected geometrically reduced case of 3.13. Replace S by this cover. The fibres are now geometrically connected and smooth, hence irreducible. If the generic constants degree was d and the genus over them g, its original arithmetic genus was $1 + d(g-1) \geq 1$, so $g \geq 1$. The fibre genus stays equal to g by flat projectivity.

Identify the function fields via the generic isomorphism. For the DVR valuation of the generic point of a closed fibre Y_s, properness gives a center on X. If its center were a closed point, its residue field has transcendence degree one over that point's residue field. The arguments of 3.14(a),(b), which require only discreteness and the dimension inequality, produce after finitely many blow-ups a codimension-one center on a regular model of X. Until that occurs, the centers are regular closed points, and all blown-up surfaces remain regular by Thm. 8.1.19.

The first divisor realizing this valuation is an exceptional projective line over a finite residue extension of $k(s)$. Its function field is therefore $k'(u)$ and is also the function field of Y_s. Because Y_s is geometrically integral, its algebraic constants equal $k(s)$; hence $k' = k(s)$. Normal projective curves with the same function field are isomorphic, so Y_s would be a projective line, contradicting $g \geq 1$. Thus every such valuation already has codimension-one center on X. The horizontal codimension-one centers agree by the generic isomorphism. Thm. 8.3.20 now extends the map to all of X. Apply the same argument to its inverse; the two composites are the identity by uniqueness. This proves existence and the asserted isomorphism. □

Exercise 3.16

> **Problem.** For dominant finite type $X \to Y$ and a discrete valuation ν of $K(X)$, restrict it to ν' on $K(Y)$. (a) Bound $\operatorname{trdeg}_{k(\nu')} k(\nu)$ by $\operatorname{trdeg}_{K(Y)} K(X)$. (b) If ν is divisorial and Y universally catenary Nagata, realize ν' on a blow-up of Y.

Solution. (a) Lift an algebraically independent family of residues over $k(\nu')$ to valuation-zero elements of $K(X)$. Any polynomial relation over $K(Y)$ can have its coefficients divided by one of minimum valuation. All coefficients then lie in the restricted valuation ring, with at least one unit; reduction gives a nonzero polynomial relation on the residues, a contradiction. Thus the lifts are algebraically independent over $K(Y)$ and the bound follows, also when ν' is trivial.

(b) **Require ν' nontrivial.** Otherwise take $X = \mathbf{A}^1_k$, $Y = \operatorname{Spec} k$, and $\nu = \operatorname{ord}_0$. It is divisorial on X, but its restriction to k is trivial and is not a codimension-one valuation on any blow-up of a point.

With this correction, choose the codimension-one normal center x of ν and put $y = f(x)$, $d = \operatorname{trdeg}_{K(Y)} K(X)$. Universal catenarity gives

$$1 + \operatorname{trdeg}_{k(y)} k(\nu) = \dim \mathcal{O}_{Y,y} + d.$$

Part (a) and the valuation dimension inequality force

$$\operatorname{trdeg}_{k(y)} k(\nu') = \dim \mathcal{O}_{Y,y} - 1.$$

The residue field is finitely generated, being an intermediate field in a finitely generated field extension. The nontrivial restriction of a discrete valuation is discrete. Apply 3.14 on an affine neighborhood of y to realize it as a DVR center after finitely many blow-ups. Their composite on that affine is a projective birational morphism, hence a single blow-up by Thm. 8.1.24. Extend its coherent center ideal to Y by scheme-theoretic closure. The resulting blow-up has the same local model and realizes ν' as claimed. □

Exercise 3.17

> **Problem.** For proper birational $f : X \to Y$ of normal fibered surfaces over an affine Dedekind base, with regular Y, (a) describe its exceptional curves; (b) find an effective Cartier D on Y with affine complement containing their images; (c) deduce f is their contraction and is projective.

Solution. The generic-fibre map is an isomorphism of normal projective curves. Thus the exceptional locus is vertical. The purity theorem for birational morphisms with regular target (Thm. 7.2.22) makes it a finite union of prime divisors E_i, proving (a).

(b) Embed Y in projective space over the affine base. Homogeneous hypersurface avoidance gives a positive-degree section nonvanishing at the finitely many points $f(E_i)$ and not vanishing identically on Y. Its zero scheme on integral Y is an effective Cartier divisor D. Its complement is closed in the affine principal projective open, hence affine. It contains the specified points.

(c) Replace D by a positive multiple so its line bundle is generated by global sections and has positive generic-fibre degree. Its pullback is trivial on every E_i, since its support avoids those

curves. On any other vertical curve its degree is positive: the image curve cannot be contained in the affine complement of D, since it is a positive-dimensional proper curve; alternatively the chosen hypersurface bundle is relatively ample. Proposition 8.3.30 constructs the projective contraction of exactly these curves. Its uniqueness, or the valuation characterization 8.3.23, identifies that contraction with f. Hence f is projective, even though only properness was initially assumed. □

Exercise 3.18

> **Problem.** Suppose projective birational $f : X \to Y$, with integral X and normal Y, contracts a connected fibre F to y_0 and is an isomorphism off it. For dominant $g : X \to Z$ constant on F, (a) prove $f_*\mathcal{O}_X = \mathcal{O}_Y$ and $\mathcal{O}_{Y,y_0} = \bigcap_{x\in F} \mathcal{O}_{X,x}$; (b) descend g through f.

Solution. The standard normal-target proper birational theorem gives $f_*\mathcal{O}_X = \mathcal{O}_Y$. A rational function regular at every point of the quasi-compact fibre is regular on a neighborhood of it. The complement of that neighborhood has closed image under f missing y_0. Thus it is regular on the inverse image of a neighborhood of y_0, and the pushforward identity makes it a germ in $\mathcal{O}_{Y,y_0}$. This proves the intersection identity; its other inclusion is immediate.

Dominance identifies $K(Z) \subseteq K(X)$. Every element of $\mathcal{O}_{Z,z_0}$ pulls back regularly at every $x \in F$, hence lies in that intersection. To see global descent without an unnecessary finite-type hypothesis on Z, first define h on points by the given map off y_0 and by $h(y_0) = z_0$. The proper surjective map f is a topological quotient, so h is continuous because $g = hf$ and g is constant on each fibre. For each open $V \subset Z$, the map on functions lands in

$$\mathcal{O}_X(g^{-1}V) = \mathcal{O}_X(f^{-1}h^{-1}V) = \mathcal{O}_Y(h^{-1}V).$$

These maps respect restrictions. They are local on stalks: a unit at y pulls back to a unit at every point above it, so the local maps inherited from g cannot send a nonunit in $\mathcal{O}_{Z,h(y)}$ to a unit at y. Thus h is a scheme morphism and $g = hf$. Surjectivity and the pushforward identity also prove uniqueness. □

Exercise 3.19

> **Problem.** For finitely many selected vertical curves on finitely many closed fibres, prove existence of their contraction equivalent to existence after localizing the base at each of those fibres.

Solution. A global projective contraction stays a contraction after localization, by the fibre and valuation description. Conversely start with a contraction over $\mathcal{O}_{S,s_i}$. Its projective equations, morphism and inverse off the exceptional set involve finitely many coefficients, so descend to an open neighborhood V_i of s_i. One can take the normal Stein target of the descended map, using normality of the source. Remove the finitely many other fibres on which the descended birational map fails to be an isomorphism; none is the distinguished fibre, since its local model is the prescribed contraction. This gives a contraction over V_i which is an isomorphism off the chosen curves.

On overlaps not containing the distinguished fibres the targets identify with the source. On overlaps which do contain one, uniqueness of contractions gives the same identification, compatible on triple intersections by uniqueness. Glue these targets with X over the complement of the finitely many fibres. Projectivity is local on the target, and the local projective contractions glue to the required contraction. This is also the usual finite-presentation spreading argument for the criterion in Prop. 8.3.30; its finiteness is why only finitely many selected fibres are involved. □

Exercise 3.20

> **Problem.** For a proper normal surface over a field and an effective Cartier D trivial on selected curves E_i and of positive degree on every other prime curve, examine the claimed eventual global generation of $\mathcal{O}(nD)$ and existence of a contraction.

Solution. The printed hypotheses do not suffice. Work over an algebraically closed field of characteristic zero. Let C be an elliptic curve and $0 \to \mathcal{O}_C \to \mathcal{V} \to \mathcal{O}_C \to 0$ a nonsplit extension. On the smooth projective ruled surface $X = \mathbf{P}_C(\mathcal{V})$ let E be the section from the quotient to $\mathcal{O}_C$. Then

$$\mathcal{O}_X(E) = \mathcal{O}_{\mathbf{P}(\mathcal{V})}(1), \qquad E^2 = 0, \qquad \mathcal{O}_X(E)|_E = \mathcal{O}_E.$$

For every $n \geq 1$, $h^0(X, \mathcal{O}_X(nE)) = h^0(C, \operatorname{Sym}^n \mathcal{V}) = 1$, with only the section vanishing on nE. To verify this calculation, choose triangular transition matrices for the extension with nonzero class $c \in H^1(C, \mathcal{O}_C)$. A hypothetical additional section has a highest nonconstant symmetric-power coefficient a. The next coefficient's gluing equation would make jac a coboundary for a positive integer j, impossible in characteristic zero. Thus only the pure power of the distinguished subline survives.

Every other prime curve Γ has $E \cdot \Gamma > 0$. Indeed write its line bundle as $\mathcal{O}_X(aE) \otimes p^*M$; its intersection with E is $\deg M \geq 0$ by effectiveness and $\Gamma \neq E$. If this degree were zero, the filtration of $\operatorname{Sym}^a \mathcal{V}$ with trivial quotients implies no sections after twisting by a nontrivial degree-zero M, and the untwisted calculation gives only the divisor aE. Both contradict $\Gamma \neq E$. Vertical fibres also have intersection one. Thus $D = E$ satisfies exactly the printed assumptions, but none of its positive powers is globally generated. Moreover E cannot be an exceptional curve of a birational contraction of a smooth surface to a normal surface, since such a curve has negative self-intersection, whereas $E^2 = 0$.

A useful valid version assumes X projective and $\operatorname{Supp} D$ disjoint from the E_i. Then $\mathcal{O}(D)|_D$ is ample by the positive-degree criterion on its component curves. Serre vanishing on D makes $H^1(D, \mathcal{O}_D(nD)) = 0$ for large n. The restriction sequences make $h^1(X, \mathcal{O}_X(nD))$ eventually decreasing and therefore stationary. They then give surjectivity of sections onto D, proving eventual global generation as in Lemma 8.3.32. Also $D^2 > 0$, so the resulting morphism has two-dimensional image. Its normal Stein factor is birational to X and contracts exactly the E_i. Since X is projective over the field, this morphism is projective. This stronger disjoint-support condition avoids the ruled-surface obstruction. □

Exercise 3.21

> **Problem.** Over a Nagata DVR R, realize finitely many divisorial valuations of $K(X)$ on a model dominating X; if R is Henselian, obtain a birational model whose special-fibre valuations are exactly the prescribed ones.

Solution. The valuations in the last assertion must be vertical, that is, nontrivial on $K(R)$, and the finite set must be nonempty; a horizontal valuation cannot become a special-fibre component, and a proper flat model cannot have an empty special fibre. Interpret "R-valuations" here in this vertical sense.

For (a), each divisorial valuation is realized on a projective birational model by the blow-up construction in Thm. 8.3.26, or by its definition as divisorial on a birational model. Normalize the closure of the graph joining finitely many such models and X. Normalization is finite because the base is Nagata and all models are finite type. This gives a normal projective model $X_1 \to X$ simultaneously dominating them. A divisor already realized remains a codimension-one valuation on a proper birational normal model, by the DVR criterion. Thus all prescribed valuations occur in the special fibre of X_1.

For (b), keep the distinct prescribed components. By Theorem 8.3.36, any proper subset of the components of a normal fibre can be contracted. Contract all components other than the selected nonempty set. The resulting normal fibered surface X_2 has exactly the prescribed divisorial valuations as its special-fibre generic points. It is birational to X; no morphism $X_2 \to X$ is asserted in this second step. □

Exercise 3.22

> **Problem.** For excellent Noetherian integral one-dimensional X, show resolution by repeated blow-ups of the reduced singular locus terminates.

Solution. Excellence makes the normalization finite and the singular locus closed, hence a finite set of closed points. A blow-up in that reduced finite set is proper birational of one-dimensional integral schemes, hence quasi-finite and finite. The normalization dominates it, because every nonzero ideal in a DVR is invertible. All successive rings therefore lie between $\mathcal{O}_X$ and its finite normalization. The quotient is a finite module supported on finitely many closed points, hence has finite length. Each nontrivial blow-up strictly enlarges the intermediate ring and consumes some of that finite length.

Thus the sequence stops. At a purported singular point of the terminal curve its reduced center ideal would be its maximal ideal; if its blow-up were an isomorphism this ideal would already be invertible. A one-dimensional local domain with principal maximal ideal is regular, a contradiction. Hence the terminal curve is regular. The centers at every step are finite reduced unions of fields, regular as schemes, and the construction is an isomorphism over the original regular locus. □

Exercise 3.23

> **Problem.** For $X = \operatorname{Proj} \mathbf{Z}[x, y, z]/(3x^3 + 4y^3 + 5z^3)$: (a) prove the fibres away from $2, 3, 5$ elliptic; (b) describe the fibre at 5 and regularity of its total-space points; (c) resolve the singularity at 2 by two rounds of point blow-ups; (d) compute three rounds at 3 and show exactly one singular point remains.

Solution. (a) The partial derivatives $9x^2, 12y^2, 15z^2$ have no common projective zero away from $2, 3, 5$. These fibres are smooth plane cubics, hence geometrically connected of genus one. Over a finite field they have a rational point: the Hasse–Weil bound gives $\#X(\mathbf{F}_p) \geq p+1-2\sqrt{p} > 0$. They are therefore elliptic curves after choosing such a point.

(b) Modulo 5 the equation is $3x^3 - y^3$, whose factors are, up to a nonzero scalar,

$$(y - 2x)(y^2 + 2xy + 4x^2).$$

The quadratic has nonsquare discriminant 3 in $\mathbf{F}_5$, so it is irreducible over that field, and defines a singular projective conic (two conjugate lines over $\mathbf{F}_{25}$) with vertex $[0 : 0 : 1]$. The other component is the line $y = 2x$. Away from the vertex a fibre partial derivative is nonzero. At the vertex use $z = 1$: $3x^3 + 4y^3 + 5$ has nonzero linear parameter 5 in the ambient maximal ideal modulo its square. Thus the total surface is regular there too, proving all of $X_5 \subset \operatorname{Reg} X$.

(c) Modulo 2 the equation is $x^3 + z^3$. Its only total-space singular point is $q_2 = [0 : 1 : 0]$; off this vertex one of the fibre partial derivatives is nonzero. Near it put $y = 1$, so the equation is $3x^3+4+5z^3$. Blow up $(2, x, z)$. The 2-chart has equation $1+6a^3+10b^3 = 0$, with $x = 2a, z = 2b$, and no exceptional special-fibre point. On the x-chart put $2 = xa, z = xb$. After saturation the equations are

$$2 = xa, \qquad a^2 + x(3 + 5b^3) = 0.$$

The exceptional reduced curve is $x = a = 0$. It is regular away from the points where $b^3 = 1$ in characteristic two: one rational point $b = 1$ and one degree-two point $b^2 + b + 1 = 0$. The z-chart is symmetric and adds no other singular point. At either bad point $c = 3 + 5b^3$ is a local parameter, since its derivative is a unit. The surface's tangent cone in the regular ambient parameters is

$$a^2 + xc = 0.$$

This is a smooth projective conic, also in characteristic two; its projective gradient has no zero on the conic. Blowing up that point therefore makes the strict transform regular along the exceptional divisor. Blow up the two disjoint points together in the second round. The resulting surface is regular. The first exceptional curve has multiplicity three, since generically $x = -a^2/(3+5b^3)$ and $2 = xa$. The two new exceptional curves have multiplicity two, the order of $2 = xa$ at their centers, and residue fields $\mathbf{F}_2, \mathbf{F}_4$. This recovers the components described in Figure 12.

(d) The fibre at 3 is the triple line $y = z$. The only singular point of the total space on it is $q_3 = [0 : 1 : 1]$. With $z = 1$, $u = x$, $v = y - 1$, the equation is

$$F = 3u^3 + 9 + 12v + 12v^2 + 4v^3 = 0.$$

At $v = 0$ its linear coefficient in the base parameter is u^3 modulo 3, so only $u = 0$ is singular. The following are saturated strict-transform equations on the charts containing the successive

singular points. Each displayed base relation defines a regular three-dimensional ambient local ring.

First blow up $(3, u, v)$ and use $3 = ua, v = ub$:

$$G = a^2 + 4ab + 4uab^2 + 4ub^3 + u^2a = 0, \qquad 3 = ua.$$

Its exceptional tangent equation is $a(a + b) = 0$ modulo 3. The unique unresolved point is $u = a = b = 0$. The 3-chart has equation $1 + 4b + 12b^2 + 12b^3 + 9a^3 = 0$ and is regular on its exceptional part; the remaining chart is regular off the same point.

Blow up this point and use $a = uc, b = ud$:

$$H = c^2 + 4cd + uc + 4u^2d^2(c + d) = 0, \qquad 3 = u^2c.$$

Again the only unresolved point is $u = c = d = 0$. For completeness the other two charts have exceptional equations $1 + 4f = 0$ and $f(f + 4) = 0$ with nonzero derivative in f; hence they introduce no further singularities.

Blow up that point and put $c = ue, d = uf$:

$$J = e^2 + 4ef + e + 4u^3f^2(e + f) = 0, \qquad 3 = u^3e.$$

On $u = 0$ its two exceptional lines meet only at $e = 0$, $f = -1/4$ modulo 3. Away from their meeting point the projective tangent equation is regular, so all other exceptional points and charts are regular. At the meeting point use $w = f + 1/4$. The equation has leading part $e^2 + 4ew$ and higher terms divisible by u^3, hence lies in the square of the ambient maximal ideal: this point is indeed still singular. Thus exactly one singular point remains after the three rounds, as requested. The generic multiplicities of the two new exceptional components are six and three, read from $3 = u^3e$ and the equation for J. No claim that three rounds complete the resolution at 3 is made. □

Exercise 3.24

> **Problem.** For a normal fibered surface over a DVR R with smooth generic fibre, prove normality after completion of the base.

Solution. Write $\widehat{X} = X \times_R \widehat{R}$. Its generic fibre is smooth over $\widehat{K}$, hence regular. Along the special fibre the completed local rings of $\widehat{X}$ and X agree by Lemma 8.3.49(b). The local rings of normal X have (S_2) and dimension at most two, hence are Cohen–Macaulay; this property is preserved and detected by completion. Thus the local rings of $\widehat{X}$ along the special fibre satisfy (S_2). Its height-one special-fibre local rings have the same completion as the corresponding DVRs on X, so are regular. The generic-fibre local rings are already regular. Serre's criterion now gives normality of $\widehat{X}$ in the local-ring sense.

If normality is taken to include irreducibility, as in Liu's simplifying convention, add geometric integrality of the generic fibre to conclude that $\widehat{X}$ is integral. Without it, the completion may split into disjoint normal components, for example for a projective line over a finite etale constants extension which splits after completing R. The local normality statement above is what the Serre-criterion proof establishes in full generality. □

Exercise 3.25

Problem. If a reduced locally Noetherian scheme has a desingularization in the strong sense, prove its regular locus open.

Solution. Let $\pi : Z \to X$ be that proper birational morphism with regular Z. Its isomorphism locus on the base is open: an isomorphism after localization at a point descends to a neighborhood by finite presentation, including its inverse; alternatively use the proper quasi-finite locus and the coherent pushforward algebra. Above each regular point it is an isomorphism by the strong-resolution condition, so that point belongs to this open. Conversely on any such open X is isomorphic to an open of regular Z and is regular. Thus the isomorphism locus is exactly $\operatorname{Reg} X$, proving openness. This does not assume in advance that the regular locus of every Noetherian scheme is open. □

Exercise 3.26

Problem. For a fibered surface over a one-dimensional Dedekind base with smooth generic fibre and any finite field extension $L/K(X)$, assess the claimed finiteness of normalization in L.

Solution. The arbitrary inseparable version is false over a non-Nagata DVR. Take a DVR R with a finite inseparable extension L_0/K inside $\widehat{K}$ as in 3.28. Its integral closure R' in L_0 is not finite by 3.28(a). Let $X = \mathbf{P}^1_R$, with smooth generic fibre, and let $L = L_0(T)$. The normalization of X in L is $\mathbf{P}^1_{R'}$: on each affine chart the integral closure is $R'[T]$, which is integral over $R[T]$, integrally closed, and has fraction field L. If this normalization were finite, quotienting its affine chart algebra by T would make R' finite over R, a contradiction.

There are two useful correct versions. If the base is Nagata, all finite type models are Nagata and normalization in every finite field extension is finite. If $L/K(X)$ is separable, first normalize X itself; that normalization is finite by the smooth-generic-fibre case of Lemma 8.3.49 and Thm. 8.3.50. On a normal Noetherian affine ring the trace-pairing lattice argument used in 2.16 proves finiteness of the integral closure in a finite separable extension. Compose the two finite maps. Applying the completion argument to the unrestricted inseparable case would be invalid: its completed generic algebra can be nonreduced, exactly the obstruction in 3.28. □

Exercise 3.27

Problem. For the last regular model X_n in the normalization/point-blow-up sequence (3.11), assess the assertion that every regular proper birational $Z \to X$ factors through X_n, and explain the valid common-refinement statement.

Solution. The specified resolution is not a universal minimal resolution. Here is a counterexample over the excellent DVR $R = k[t]_{(t)}$, with algebraically closed k of characteristic zero. Let E be the smooth cubic $V^2W = U^3 + W^3$, with $o = [0 : 1 : 0]$, and put

$$X = V_+(Y^2Z - X_0^3 - t^6Z^3) \subset \mathbf{P}^2_R.$$

The only total-space singular point is $q = (t, X_0/Z, Y/Z) = (0, 0, 0)$; the generic fibre is smooth, and the hypersurface is normal by (S_2) and regularity in codimension one.

The generic isomorphism $E_R \dashrightarrow X$ is given by $[U : V : W] \mapsto [t^2U : t^3V : W]$. Blow up E_R at $(o, t = 0)$, obtaining regular Z. This resolves that rational map: near o write $q_0 = U/V$, $r = W/V$, so $r = q_0^3 + r^3 = q_0^3 h$ with h a unit. Its base ideal is $(t^2 q_0, t^3, q_0^3 h)$, which becomes principal on both charts of the blow-up of (t, q_0). Thus it gives a proper birational morphism $Z \to X$.

On the chart $t = q_0 a$ the image near the target cusp is

$$X_0/Z = a^2/h, \qquad Y/Z = a^3/h, \qquad t = q_0 a.$$

The pullback of the maximal ideal of q is therefore $a(q_0, a)$ at $q_0 = a = 0$, not invertible in this regular two-dimensional local ring. By the blow-up universal property Z does not even map to $\mathrm{Bl}_q X$. Since the canonical sequence for normal singular X passes through that blow-up and its normalization, Z cannot factor through its terminal X_n. This disproves (a) and the factorization assumed in the wording of (b).

Projectivity of $Z \to X$ nevertheless holds over the affine Dedekind base: regular proper flat Z/S is projective by Lichtenbaum's theorem 8.3.16, and its graph gives a projective morphism to separated X. The valid replacement for (b) is a *common* regular refinement W of Z and X_n, where $W \to X_n$ is a succession of blow-ups at finite regular point centers. This is elimination of indeterminacy for birational maps of regular surfaces, or the factorization theorem of Chapter 9. It does not produce a morphism $Z \to X_n$. The abstract minimal resolution of a normal surface singularity is a different construction from the particular sequence (3.11). □

Exercise 3.28

> **Problem.** Let a non-excellent DVR field K have a finite inseparable extension $L \subset \widehat{K}$. (a) Show its integral closure is not finite over $R = \mathcal{O}_K$. (b) Spread $\mathbf{P}^1_L$ to an integral projective flat R-scheme. (c) Show it has no alteration by a regular scheme.

Solution. (a) The finite $\widehat{K}$-algebra $L \otimes_K \widehat{K}$ is nonreduced. Indeed the embedding $L \hookrightarrow \widehat{K}$ selects a factor of the separable part, and the remaining nontrivial inseparable minimal polynomial acquires a repeated root in that factor. If the integral closure R' were finite over R, it would be a semilocal normal Noetherian one-dimensional ring. Its t-adic completion would be a product of completed DVRs, hence reduced, and inverting t would identify it with $L \otimes_K \widehat{K}$, a contradiction. This argument also covers a mixed separable/inseparable finite extension, not only a purely inseparable one.

(b) The curve $\mathbf{P}^1_L$, considered over K, is projective. Choose a projective embedding and take its scheme-theoretic closure in projective space over R. That closure is integral, its homogeneous coordinate algebra is R-torsion-free, and therefore it is flat over the DVR. Its generic fibre is the given curve, giving the required model X.

(c) Suppose a regular alteration $Y \to X$ existed. Then Y/R is integral, proper, flat and two-dimensional, with regular generic fibre. Lichtenbaum makes it projective. Lemma 8.3.49(c), applied to its identity resolution, gives regular $Y_{\widehat{K}}$. The generic map embeds L into $H^0(Y_K, \mathcal{O})$;

flat base change gives an injection

$$L \otimes_K \widehat{K} \hookrightarrow H^0(Y_{\widehat{K}}, \mathcal{O}).$$

Its source has nonzero nilpotents by (a), while its target is reduced by regularity. This is impossible, so no regular alteration exists. □